

SOC-5811 Week 6: Simulation

Nick Graetz

University of Minnesota, Department of Sociology

10/6/2025



REVIEW

- ▶ Sampling distributions



REVIEW

- ▶ Sampling distributions
- ▶ Statistical inference and p-values



REVIEW

- ▶ **Standard deviation** = a measure of the spread of data around the sample mean.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

- ▶ **Standard error** = a measure of the spread of sample means around the population mean; the standard deviation of the sampling distribution.

$$\text{Standard error} = \frac{\sigma}{\sqrt{n}}$$

SIMULATION

- ▶ Last week was **statistical inference**: We *observe* sample data $\rightarrow$ we estimate the most likely *unknown* population model that produced that sample data.



SIMULATION

- ▶ Last week was **statistical inference**: We *observe* sample data → we estimate the most likely *unknown* population model that produced that sample data.
- ▶ This week is **simulation**: We *know* the population model → we simulate all the sample data is could plausibly produce (i.e., we simulate the *sampling distribution*).



SIMULATION

Why is simulation an important concept/skill?

- ▶ Develop statistical intuition without needing math.



SIMULATION

Why is simulation an important concept/skill?

- ▶ Develop statistical intuition without needing math.
- ▶ We can use simulation to approximate the sampling distribution of data and propagate this to the sampling distribution of statistical estimates (aka **bootstrapping**).



SIMULATION

Why is simulation an important concept/skill?

- ▶ Develop statistical intuition without needing math.
- ▶ We can use simulation to approximate the sampling distribution of data and propagate this to the sampling distribution of statistical estimates (aka **bootstrapping**).
- ▶ Very important principle for more complex models: you can always simulate.



SIMULATION

- ▶ Remember: this is all about understanding how we do **statistical inference**.



SIMULATION

- ▶ Remember: this is all about understanding how we do **statistical inference**.
- ▶ What can I learn about the population model from the data I observe?



SIMULATION

- ▶ Remember: this is all about understanding how we do **statistical inference**.
- ▶ What can I learn about the population model from the data I observe?
- ▶ Are my sample parameters real, or could they plausibly have been observed by chance?



SIMULATION

Table 2. Children with Same-Sex Parents and Standardized Test Scores at the End of Primary Education

	All Children		Children Raised by Same-Sex Parents from Birth ^a		
	(1)	(2)	(3)	(4)	(5)
Child has same-sex parents (1 is yes)	.106*** (.019)	.054** (.018)	.104*** (.024)	.139*** (.023)	.147*** (.041)
Sex (1 is male)	-.004* (.002)	-.009*** (.002)	-.003 (.002)	-.007*** (.002)	
Ethnicity (ref: both parents born in NL)					
One parent NL, other Western country		.004 (.003)		.004 (.003)	
One parent NL, other non-Western country		-.031*** (.005)		-.031*** (.005)	
Both parents not NL		-.111*** (.006)		-.111*** (.006)	
Parental education (1 is diploma SE)		.531*** (.004)		.532*** (.004)	
Log household income		.083*** (.002)		.084*** (.002)	
Mean parental age		.018*** (.000)		.018*** (.000)	
Family size		-.066*** (.001)		-.066*** (.001)	
Family structure (ref: married parents)					
Cohabiting parents		-.073*** (.002)		-.073*** (.002)	
Other		-.367*** (.012)		-.154*** (.032)	
Fixed effects					
Birth year	Yes	Yes	Yes	Yes	
Neighborhood	No	Yes	No	Yes	
Method	OLS	OLS	OLS	OLS	CEM ^b
Number of children	1,204,692	1,204,692	1,195,624	1,195,624	48,388
Number of children with same-sex parents	2,971	2,971	1,390	1,390	1,390
R ²	.004	.108	.004	.109	.121

SIMULATION

Table 1

Estimates of the Effect of Veteran Status on Homeownership by Race (1960)

	Outcome	All	White	Black
Stage I	Veteran status	.396*** (.005)	.409*** (.005)	.274*** (.009)
Reduced Form	Homeownership	.048*** (.003)	.051*** (.003)	.020** (.005)
IV	Homeownership	.121*** (.008)	.125*** (.009)	.071*** (.019)
OLS	Homeownership	.114*** (.006)	.104*** (.007)	.080*** (.011)

THIS WEEK: R SKILLS

► Functions in R



THIS WEEK: R SKILLS

- ▶ Functions in R
- ▶ Simulating in R



THIS WEEK: R SKILLS

- ▶ Functions in R
- ▶ Simulating in R
- ▶ Bootstrapping in R



THIS WEEK: R SKILLS

- ▶ Functions in R
- ▶ Simulating in R
- ▶ Bootstrapping in R
- ▶ Introduction to multivariable models



FUNCTIONS IN R

```
log(5)
```

```
## [1] 1.609438
```

FUNCTIONS IN R

```
new_function <- function(x) {  
  y <- x*10  
  return(y)  
}
```

FUNCTIONS IN R

```
new_function <- function(x) {  
  y <- x*10  
  return(y)  
}  
new_function(5)
```

```
## [1] 50
```

FUNCTIONS IN R

```
new_function <- function(x) {  
  y <- x*10  
  return(y)  
}  
sapply(c(5,10,15,20), new_function)
```

```
## [1] 50 100 150 200
```

SIMULATION

- ▶ Random variables



SIMULATION

- ▶ Random variables
- ▶ Distributions:



SIMULATION

- ▶ Random variables
- ▶ Distributions:
 - ▶ Normal distribution

SIMULATION

- ▶ Random variables
- ▶ Distributions:
 - ▶ Normal distribution
 - ▶ Binomial distribution



SIMULATION

- ▶ Random variables
- ▶ Distributions:
 - ▶ Normal distribution
 - ▶ Binomial distribution
- ▶ **Simulation of random variables in R.**

SIMULATION

X = the number of heads I get if I flip a coin 100 times.

X = Binomial($n=100$, $p=0.5$)

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 46
```

SIMULATION

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 56
```

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 38
```

```
rbinom(n=1, size=100, prob=0.5)
```

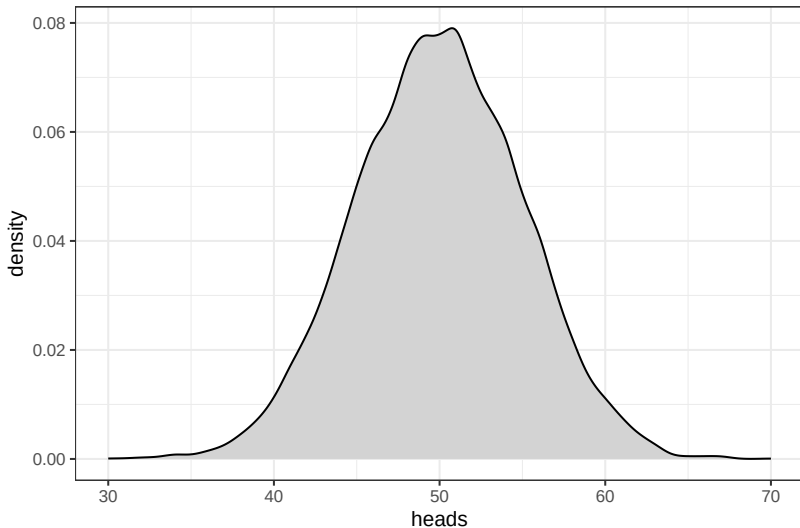
```
## [1] 61
```

SIMULATION

```
trials <- rbinom(n=10000, size=100, prob=0.5)
trials <- data.table(heads=trials)
head(trials)
```

```
##      heads
##      <int>
## 1:      53
## 2:      47
## 3:      52
## 4:      51
## 5:      44
## 6:      65
```

SIMULATION



18/59

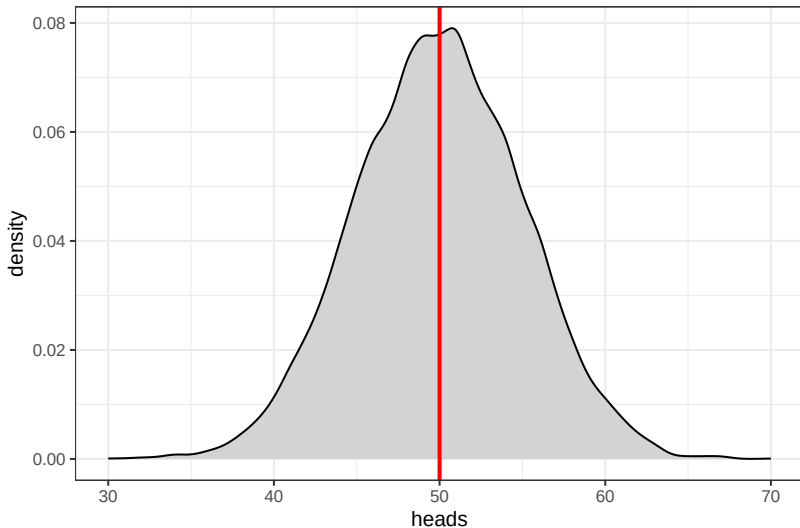


UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



SIMULATION



19/59



UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



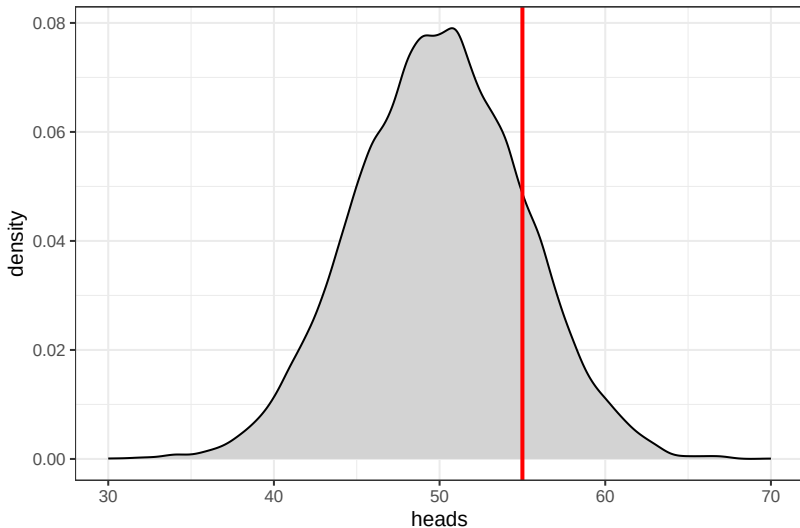
SIMULATION

Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

SIMULATION



21/59



UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



SIMULATION

```
pbinom(55, 100, 0.5, lower.tail = FALSE)
```

```
## [1] 0.1356265
```

SIMULATION

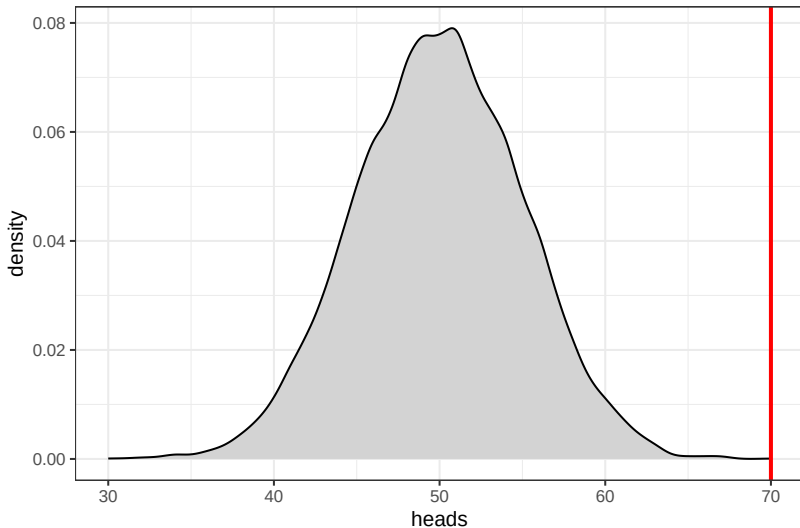
Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

Is the hypothesis that this is a fair coin consistent with our data? Could a fair coin plausibly produce our data?

SIMULATION



24/59



UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



SIMULATION

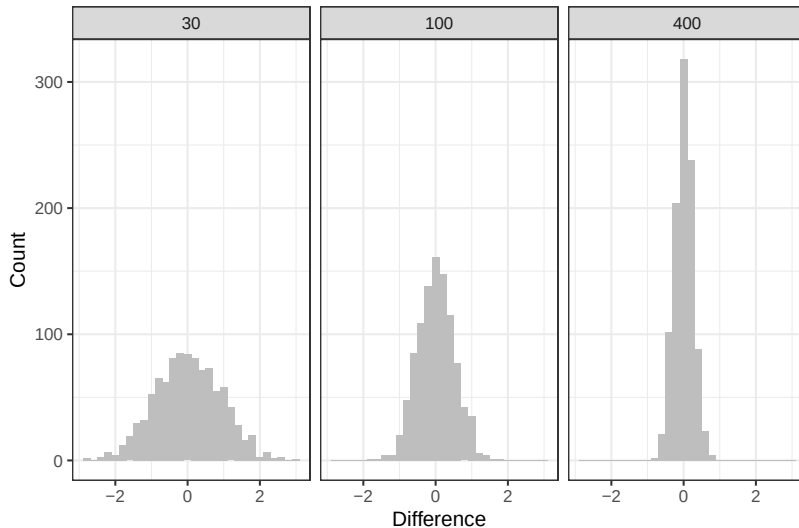
```
pbinom(70, 100, 0.5, lower.tail = FALSE)
```

```
## [1] 1.608001e-05
```

SIMULATION

```
mean_of_normal_trials <- function(i, ss) {  
  mean <- mean(rnorm(ss, 0, 5))  
  return(mean)  
}  
normal_trials <- function(ss) {  
  trials <- lapply(1:1000,  
                  mean_of_normal_trials,  
                  ss=ss)  
  trials <- data.table(sample_size=ss,  
                      mean=unlist(trials))  
  return(trials)  
}  
trials <- lapply(c(30,100,400), normal_trials)  
trials <- rbindlist(trials)
```

SIMULATION



27/59

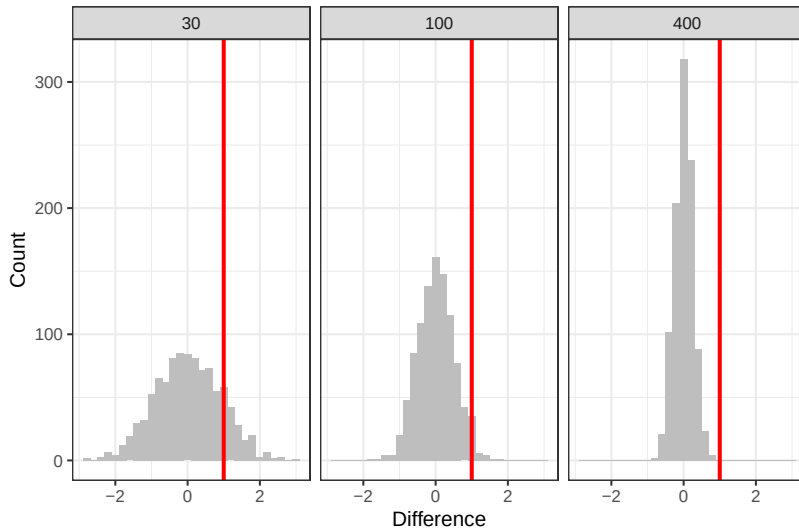


UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



SIMULATION



BOOTSTRAPPING

- ▶ Simulating a sampling distribution when you **do not know** the underlying probability model.
- ▶ Randomly sample the **observed** data and calculate your statistic many times (“bootstrapping”).



BOOTSTRAP

```
psid <- fread('https://raw.githubusercontent.com/ngraetz/soc5811/refs/heads/main/data/psid.csv')
psid <- psid %>%
  filter(g3_house_value!=0)
```

BOOTSTRAP

```
mean_analytic <- psid %>%  
  summarise(mean=mean(g3_house_value)) %>%  
  pull(mean)  
se_analytic <- psid %>%  
  summarise(se=sd(g3_house_value)/sqrt(nrow(psid))) %>%  
  pull(se)  
mean_analytic
```

```
## [1] 388711.5
```

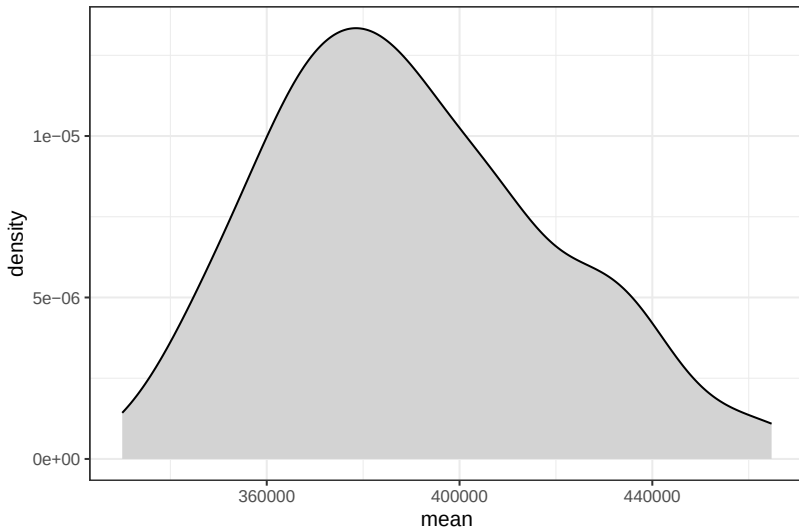
```
se_analytic
```

```
## [1] 31072.29
```

BOOTSTRAP

```
get_sample <- function(i) {  
  random_sample <- psid %>%  
    sample_n(size=nrow(psid), replace=T) %>%  
    summarise(mean=mean(g3_house_value))  
  return(random_sample)  
}  
samples <- rbindlist(lapply(1:draws, get_sample))
```

BOOTSTRAP



BOOTSTRAP

```
## Calculate standard error  
se_bootstrap <- samples %>%  
  summarise(sd=sd(mean)) %>%  
  pull(sd)
```

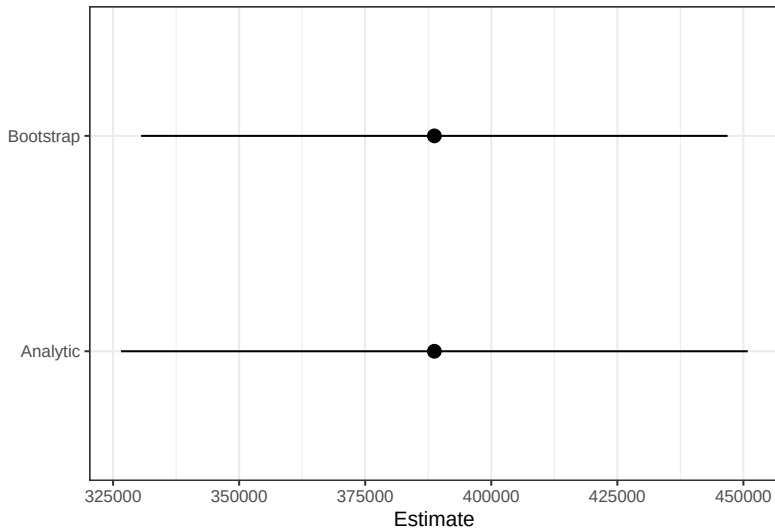
```
## Compare  
se_analytic
```

```
## [1] 31072.29
```

```
se_bootstrap
```

```
## [1] 29080.31
```

BOOTSTRAP



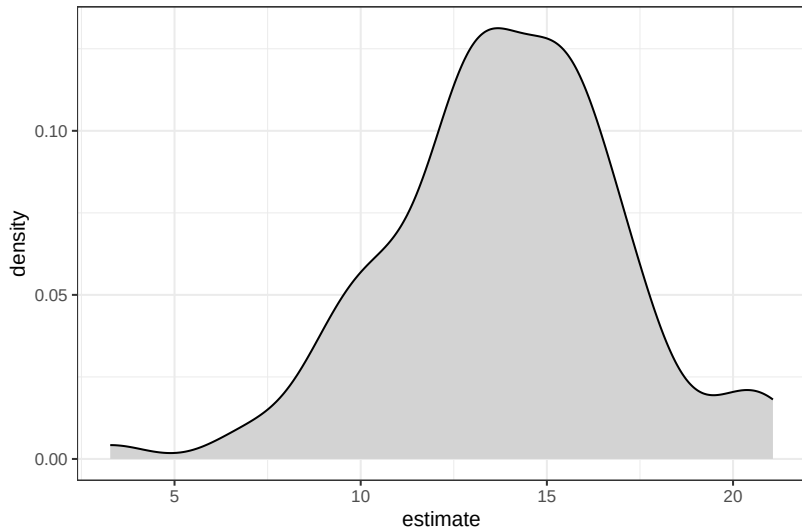
35/59



BOOTSTRAP

```
psid_subset <- psid %>%  
  filter(!is.na(g1_edu_cat) & g3_house_value!=0) %>%  
  mutate(above_hs=ifelse(g1_edu_cat %in% c('less_hs', 'hs',  
                                           'less', 'more'))  
  
get_sample <- function(i) {  
  random_sample <- psid_subset %>%  
    sample_n(size=nrow(psid_subset), replace=T)  
  model <- lm(g3_house_value_percentile~above_hs,  
              data=random_sample)  
  out <- avg_slopes(model)  
  return(out)  
}  
coef_samples <- rbindlist(lapply(1:draws, get_sample))
```

BOOTSTRAP



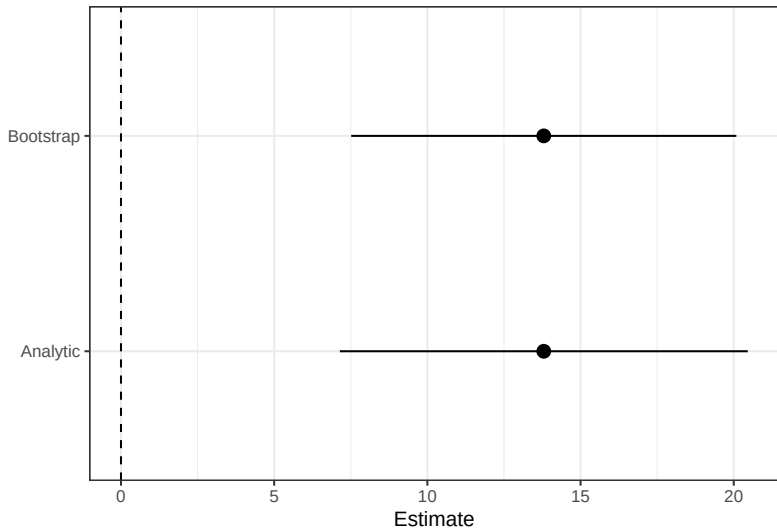
37/59



BOOTSTRAP

```
## Compare to lm()  
model <- lm(g3_house_value_percentile~above_hs,  
            data=psid_subset)  
analytic_coefs <- as.data.table(avg_slopes(model))
```

BOOTSTRAP



39/59



REVIEW

- ▶ Sampling distributions, statistical inference, and p-values.



REVIEW

- ▶ Sampling distributions, statistical inference, and p-values.
- ▶ What do we need to calculate? **Standard errors / confidence intervals.**



REVIEW

How do we calculate standard errors / confidence intervals?

- ▶ **Analytic approach:** Use formulas.



How do we calculate standard errors / confidence intervals?

- ▶ **Analytic approach:** Use formulas.
- ▶ **Simulation approach:** Look at the variation in many random samples (with replacement) from the data, which can be taken as an approximation to the sampling distribution of our data collection process; that is, possible data we could have seen, had the data collection been conducted again in the same population.

REVIEW

```
bootstrap <- function(i) {  
  1. Randomly sample with replacement from our data  
  2. Calculate our target stat.  
}  
3. Bootstrap many times to approximate the sampling  
4. Standard error = standard deviation of the sampl
```

REVIEW

Why do we do all of this?

- ▶ Estimate standard errors / confidence intervals.



REVIEW

Why do we do all of this?

- ▶ Estimate standard errors / confidence intervals.
- ▶ Estimate likelihood we may have seen a result as extreme or more extreme than our observed result if some null model were true.



REVIEW

Why do we do all of this?

- ▶ Estimate standard errors / confidence intervals.
- ▶ Estimate likelihood we may have seen a result as extreme or more extreme than our observed result if some null model were true.
- ▶ Infer something about the population model!



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?

- Predictions: few social science processes can be accounted for by a single explanatory variable (i.e., our predictions will be crude).



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?

- ▶ Predictions: few social science processes can be accounted for by a single explanatory variable (i.e., our predictions will be crude).
- ▶ Causal inference: omitted variable bias and confounding.



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?

- ▶ Predictions: few social science processes can be accounted for by a single explanatory variable (i.e., our predictions will be crude).
- ▶ Causal inference: omitted variable bias and confounding.
 - ▶ This can be an issue where we are intentionally trying to estimate a causal effect.



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?

- ▶ Predictions: few social science processes can be accounted for by a single explanatory variable (i.e., our predictions will be crude).
- ▶ Causal inference: omitted variable bias and confounding.
 - ▶ This can be an issue where we are intentionally trying to estimate a causal effect.
 - ▶ It can also be an issue where we are just trying to make a simple descriptive comparison and not thinking through exactly *which* comparison we care about.

SIMPSON'S PARADOX

Simpson's paradox

- ▶ Aggregation problem: an association may not be true for all subgroups (i.e., the problem is we are *generalizing* a high level pattern to lower levels).



SIMPSON'S PARADOX

Simpson's paradox

- ▶ Aggregation problem: an association may not be true for all subgroups (i.e., the problem is we are *generalizing* a high level pattern to lower levels).
- ▶ It's not really a "paradox" - it just depends what you're interested in!



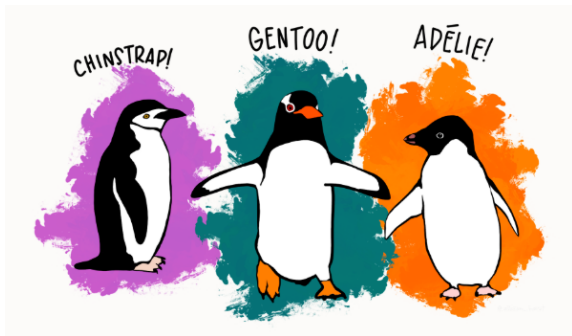
SIMPSON'S PARADOX

Simpson's paradox

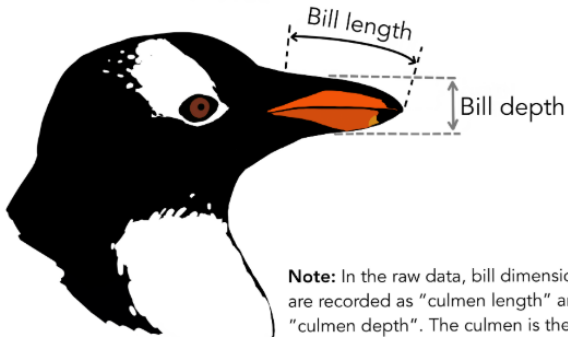
- ▶ Aggregation problem: an association may not be true for all subgroups (i.e., the problem is we are *generalizing* a high level pattern to lower levels).
- ▶ It's not really a "paradox" - it just depends what you're interested in!
- ▶ It's a useful tool for understanding why we need to be really specific about our research questions.

SIMPSON'S PARADOX

Data on 344 penguins were collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER, a member of the Long Term Ecological Research Network.



SIMPSON'S PARADOX



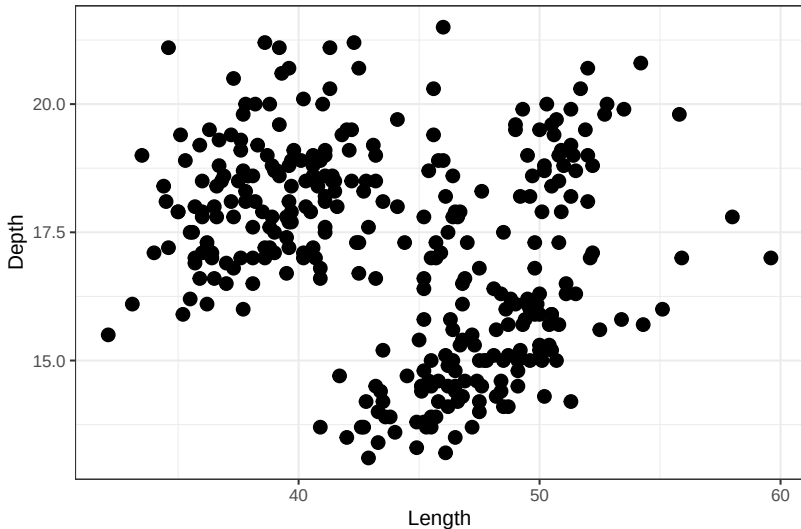
Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.

SIMPSON'S PARADOX

```
penguin_df <-  
  palmerpenguins::penguins %>%  
    na.omit()  
head(penguin_df)
```

```
## # A tibble: 6 x 8  
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g  
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>  
## 1 Adelie  Torgersen         39.1          18.7          181          3750  
## 2 Adelie  Torgersen         39.5          17.4          186          3800  
## 3 Adelie  Torgersen         40.3          18           195          3250  
## 4 Adelie  Torgersen         36.7          19.3          193          3450  
## 5 Adelie  Torgersen         39.3          20.6          190          3650  
## 6 Adelie  Torgersen         38.9          17.8          181          3625  
## # i 2 more variables: sex <fct>, year <int>
```

SIMPSON'S PARADOX



49/59

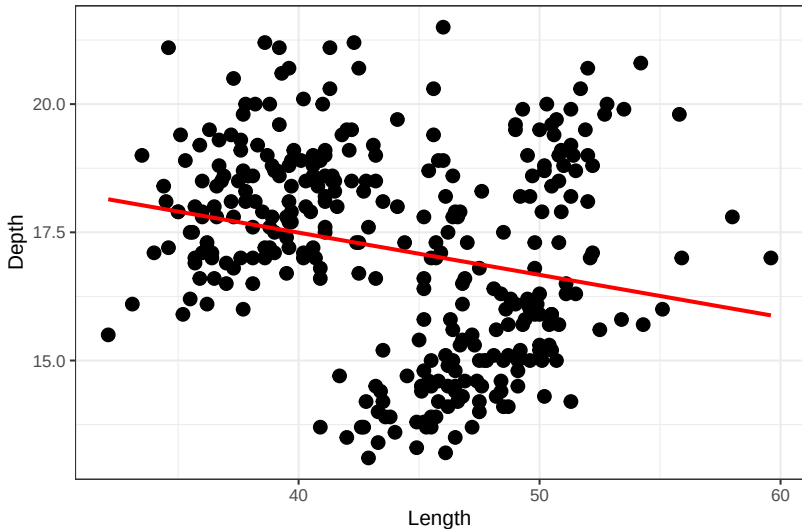


UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



SIMPSON'S PARADOX



50/59



UNIVERSITY OF MINNESOTA

Driven to DiscoverSM

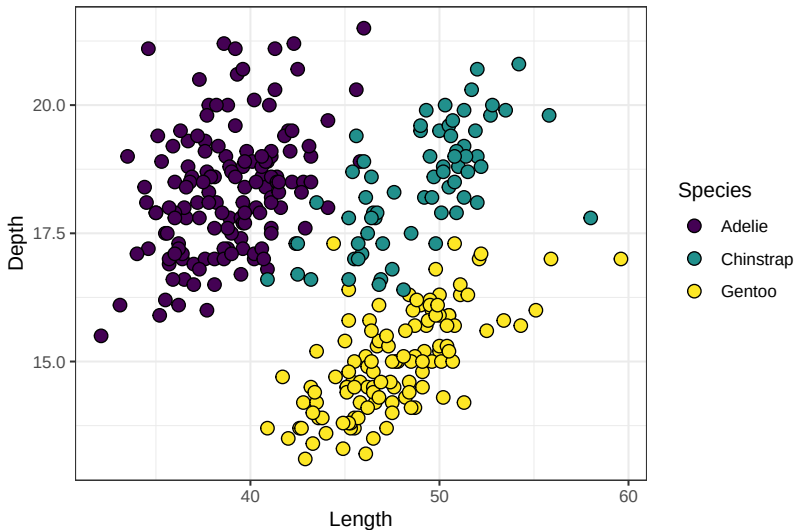


SIMPSON'S PARADOX

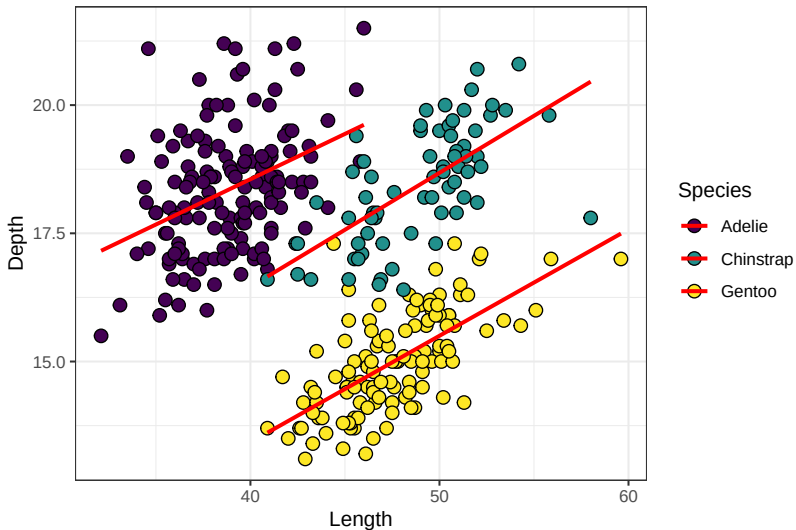
```
summary(lm(bill_depth_mm ~ bill_length_mm, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.1548 -1.4291  0.0122  1.3994  4.5004
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   20.78665    0.85417   24.335 < 2e-16 ***
## bill_length_mm -0.08233    0.01927   -4.273 2.53e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.92 on 331 degrees of freedom
## Multiple R-squared:  0.05227,    Adjusted R-squared:  0.04941
## F-statistic: 18.26 on 1 and 331 DF,  p-value: 2.528e-05
```

SIMPSON'S PARADOX



SIMPSON'S PARADOX



SIMPSON'S PARADOX

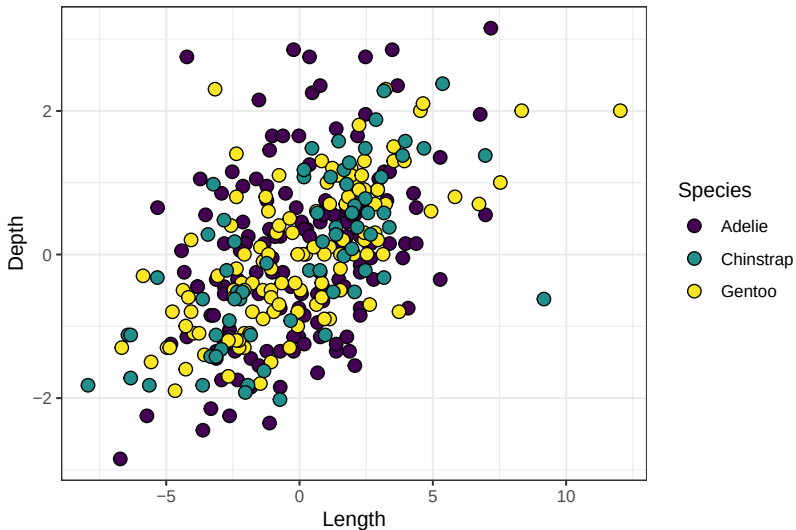
```
summary(lm(bill_depth_mm ~ bill_length_mm + species, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4579 -0.6814 -0.0431  0.5441  3.5994
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   10.56526    0.69093   15.291 < 2e-16 ***
## bill_length_mm    0.20044    0.01768   11.337 < 2e-16 ***
## speciesChinstrap -1.93308    0.22572  -8.564 4.26e-16 ***
## speciesGentoo   -5.10332    0.19440 -26.252 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9543 on 329 degrees of freedom
## Multiple R-squared:  0.7673, Adjusted R-squared:  0.7652
## F-statistic: 361.6 on 3 and 329 DF,  p-value: < 2.2e-16
```

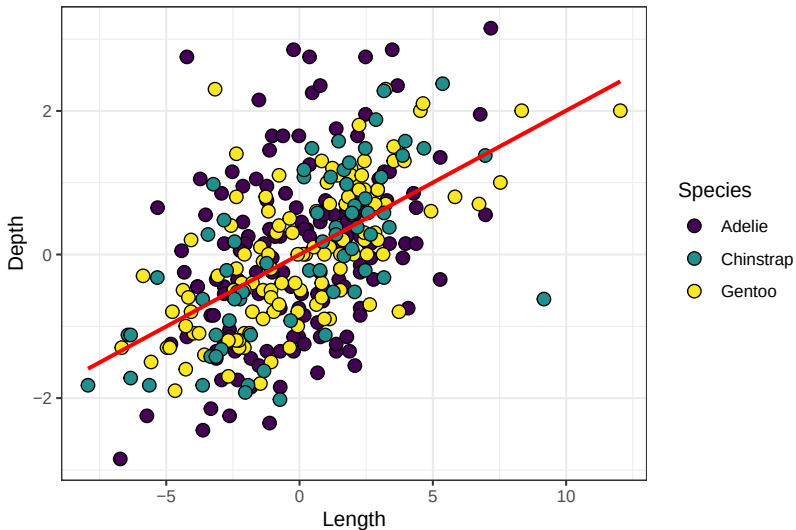
SIMPSON'S PARADOX

```
penguin_df <- penguin_df %>%  
  group_by(species) %>%  
  mutate(bill_depth_mm_demeaned = bill_depth_mm - mean(bill_depth_mm),  
         bill_length_mm_demeaned = bill_length_mm - mean(bill_length_mm))
```

SIMPSON'S PARADOX



SIMPSON'S PARADOX



57/59



UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



SIMPSON'S PARADOX

```
summary(lm(bill_depth_mm_demeaned ~ bill_length_mm_demeaned, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm_demeaned ~ bill_length_mm_demeaned,
##     data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4579 -0.6814 -0.0431  0.5441  3.5994
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.979e-16  5.214e-02   0.00    1
## bill_length_mm_demeaned 2.004e-01  1.763e-02  11.37 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9514 on 331 degrees of freedom
## Multiple R-squared:  0.2809, Adjusted R-squared:  0.2788
## F-statistic: 129.3 on 1 and 331 DF,  p-value: < 2.2e-16
```


REGRESSION WITH MULTIPLE VARIABLES

Causal inference

- ▶ People who eat more chocolate live longer.



REGRESSION WITH MULTIPLE VARIABLES

Causal inference

- ▶ People who eat more chocolate live longer.
- ▶ Key question: Are people who eat chocolate different than people who don't?